EFFICIENT ESTIMATION OF THE CONDITIONAL EFFECTS OF LONGITUDINAL MODIFIED TREATMENT POLICIES

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ABSTRACT

Longitudinal modified treatment policies (LMTPs) are an increasingly common approach for the estimation and inference of causal effects of hypothetical changes to exposure levels received over time in the presence of measured time-varying confounders. However, little work has been done on the problem of characterizing effect heterogeneity in LMTP effects across discrete subgroups or with continuous effect modifiers. We present a novel targeted learning approach for the efficient estimation of subgroup specific LMTP effects that provides multiply robust, asymptotically efficient substitution estimators. We additionally outline a procedure for simultaneous inference across multiple subgroup-specific LMTP effects. For handling the effects of continuous baseline effect modifiers, we develop a meta-learning approach for estimating conditional LMTP effects, and we present techniques for carrying out efficient estimation of collections of conditional LMTP parameters across distinct LMTPs via a working marginal structural model thereby allowing for interpretable summaries of effect heterogeneity across a range of interventions in the longitudinal setting. We illustrate our method by applying it to a question of how different deforestation policies would have differentially affected malaria incidence rates in Madagascar during the 2010-2023 timeframe at 2,773 clinics with effect heterogeneity expected by ecological zone.

Keywords: Causal inference, Longitudinal Modified Treatment Policies, Treatment Effect Heterogeneity, Efficient Inference, Subgroup Effects

1 Introduction

Longitudinal studies in medicine, epidemiology, and the environmental sciences often seek to evaluate interventions on exposures or treatments that evolve over time. In such settings, treatment decisions may depend on a subject's previous treatment and covariate history, and time-varying covariates may themselves be affected by earlier treatment. These features rule out standard regression adjustment and motivate the use of causal estimands defined through longitudinal interventions.

Longitudinal modified treatment policies (LMTPs) provide one useful class of such interventions. Rather than setting treatment to one fixed value, an LMTP modifies the treatment that a subject would otherwise have received according to a user-specified policy map. This makes LMTPs attractive in settings where deterministic static interventions are implausible or scientifically less relevant. For example, in studies of environmental exposures, one may be interested in shifting an exposure downward for only units above a threshold; and in health studies, modifying a dose or treatment intensity relative to the value actually received. LMTPs therefore provide a flexible way to encode realistic longitudinal interventions while retaining identification under standard sequential causal assumptions.

Most methodological work on LMTPs has focused on marginal causal parameters, such as the population mean outcome under a single policy or the contrast between two policies. However, scientific questions about policy effects are often inherently conditional. Investigators may ask whether an intervention has larger effects in certain ecological zones, clinical subgroups, baseline risk strata, or other baseline-defined populations. More generally, one may wish to estimate a conditional effect curve describing how the causal effect of a longitudinal policy varies with baseline covariates. Restricting attention to only marginal LMTP effects can obscure such heterogeneity.

This paper studies estimation and inference for conditional effects of LMTPs. Let V denote baseline covariates with respect to which effect heterogeneity is of interest. For an LMTP $\mathbb{d}$, we define the conditional LMTP mean

$$\theta_{\mathbb{d}}(v) = \mathbb{E}\{Y(\mathbb{d}) \mid V = v\},$$

and, for two policies $\mathbb{d}_1$ and $\mathbb{d}_0$, the conditional LMTP contrast

$$\tau(v) = \theta_{\mathbb{d}_1}(v) - \theta_{\mathbb{d}_0}(v).$$

When V is discrete, this target corresponds to a vector of subgroup-specific policy effects. When V is continuous or high-dimensional, it corresponds to a conditional effect function.

We will first present a formal identification argument for conditional LMTP means and contrasts using the observed-data likelihood factorization, the post-policy law, and the longitudinal g-formula. This yields inverse-probability and iterated-regression representations of the conditional LMTP effect. Second, for finite baseline subgroups, we develop a targeted learning estimator for the vector of subgroup LMTP means or contrasts. This estimator solves the efficient influence-function estimating equations for all subgroups jointly and supports Wald-type and simultaneous inference. Third, for general conditioning covariates V, we develop a DR-Learner for conditional LMTP effects. The DR-Learner estimates uncentered influence-function pseudo-outcomes for marginal LMTP parameters and then regresses these pseudo-outcomes on V. We establish conditions under which this estimator is asymptotically equivalent to the benchmark second-stage regression that would be obtained if the nuisance functions were known.

The remainder of the paper is organized as follows. Section 2 defines the observed data structure, LMTPs, conditional target parameters, identification assumptions, identification results and derives the pseudo-outcome representation used by the DR-Learner. Section 3 presents the finite-subgroup targeted minimum loss-based estimator and the conditional LMTP DR-Learner. Section 4 gives the large-sample properties of the proposed estimators. Section **??** presents simulation studies, and Section 6 applies the methods to estimate heterogeneous effects of deforestation on malaria incidence in Madagascar. Section 7 offers discussion of potential applications for conditional LMTP estimation and future directions for conditional effect estimation in longitudinal causal inference.

2 Setup and Target Parameters

2.1 Observed data and likelihood factorization

Let

$$O = (L_0, A_0, L_1, A_1, \ldots, L_\tau, A_\tau, Y)$$

denote the observed data for a randomly sampled unit. Here L_0 denotes baseline covariates, A_t denotes the treatment or exposure at time $t = 0, \ldots, \tau$, and L_{t+1} denotes covariates measured after treatment A_t, with $L_{\tau+1} \equiv Y$. Let

$$\overline{L}_t = (L_0, \ldots, L_t), \qquad \overline{A}_t = (A_0, \ldots, A_t),$$

and define the treatment history available immediately before treatment time t by

$$H_t = (\overline{L}_t, \overline{A}_{t-1}),$$

with the convention that $H_0 = L_0$. We assume throughout that $O_1, \ldots, O_n$ are independent and identically distributed according to an unknown law P.

Assume that P is dominated by a product measure and has density

$$p(o) = q_0(l_0) \prod_{t=0}^{\tau} g_t(a_t \mid h_t) q_{t+1}(l_{t+1} \mid h_t, a_t),$$

where

$$g_t(a_t \mid h_t) = p(A_t = a_t \mid H_t = h_t)$$

denotes the conditional density or probability mass function of the treatment at time t, and

$$q_{t+1}(l_{t+1} \mid h_t, a_t) = p(L_{t+1} = l_{t+1} \mid H_t = h_t, A_t = a_t)$$

denotes the conditional law of the next covariate, or of the final outcome when $t = \tau$. We write

$$q = (q_0, q_1, \ldots, q_{\tau+1}), \qquad g = (g_0, \ldots, g_\tau),$$

so that $p = qg$. This factorization separates the conditional treatment process g from the covariate and outcome transition laws q. The causal parameters considered below are identified by replacing the observed treatment process g with a post-policy treatment process while leaving q unchanged.

2.2 Longitudinal modified treatment policies

A longitudinal modified treatment policy is a sequence

$$\mathrm{d} = (d_0, \ldots, d_\tau)$$

of known maps. At time t, the map d_t takes an observed treatment value and history (a_t, h_t) and returns the treatment value that would be assigned under the policy,

$$a_t^{\mathrm{d}} = d_t(a_t, h_t).$$

The corresponding random variable is

$$A_t^{\mathrm{d}} = d_t(A_t, H_t).$$

For example, with a continuous exposure, d_t might shift exposure downward by a fixed amount among units above a threshold. With a discrete exposure, d_t might map some observed treatment levels to modified levels while leaving other levels unchanged.

In the longitudinal setting, the policy is evaluated recursively. Earlier modified treatments affect later histories, and later modified treatments are therefore functions of histories that include previously modified treatment values. Let

$$H_t^{\mathrm{d}} = (\overline{L}_t, A_0^{\mathrm{d}}, \ldots, A_{t-1}^{\mathrm{d}})$$

denote the history through time t under the policy. Then the policy-modified treatment process is generated recursively by

$$A_0^{\mathrm{d}} = d_0(A_0, L_0),$$

and, for $t = 1, \ldots, \tau$,

$$A_t^{\mathrm{d}} = d_t\{A_t(\overline{A}_{t-1}^{\mathrm{d}}), H_t^{\mathrm{d}}\},$$

where $A_t(\overline{A}_{t-1}^{\mathrm{d}})$ denotes the treatment value that would have been observed at time t had treatment through time $t-1$ been modified according to d.

Let

$$g_t^{\mathrm{d}}(a_t \mid h_t)$$

denote the conditional law of A_t^{d} given $H_t = h_t$. For discrete treatments, g_t^{d} is the pushforward of the observed treatment law g_t under the map d_t. For continuous treatments, we assume that $d_t(\cdot, h_t)$ is piecewise smooth and invertible on a finite partition of the treatment space, so that g_t^{d} is well-defined by the change-of-variables formula.

The post-policy distribution P^{d} is the law with density

$$p^{\mathrm{d}}(o) = q_0(l_0) \prod_{t=0}^{\tau} g_t^{\mathrm{d}}(a_t \mid h_t) q_{t+1}(l_{t+1} \mid h_t, a_t).$$

Thus the intervention changes the conditional treatment law at each time point but leaves the conditional covariate and outcome transition laws unchanged. This is the usual longitudinal g-formula representation specialized to modified treatment policies.

2.3 Conditional LMTP means and contrasts

Let $V \subseteq L_0$ denote baseline covariates with respect to which effect heterogeneity is of interest. The covariates V may be a discrete subgroup indicator, a vector of baseline features, or a lower-dimensional summary of baseline risk. Because V is measured before treatment, its distribution is not changed by the intervention.

For a single LMTP d, define the marginal LMTP mean

$$\psi_{\mathrm{d}} = \mathbb{E}_{P^{\mathrm{d}}}(Y) = \mathbb{E}\{Y(\mathrm{d})\}.$$

The corresponding conditional LMTP mean is

$$\theta_{\mathrm{d}}(v) = \mathbb{E}_{P^{\mathrm{d}}}(Y \mid V = v) = \mathbb{E}\{Y(\mathrm{d}) \mid V = v\}.$$

For two policies d_1 and d_0, define the conditional LMTP contrast

$$\tau(v) = \theta_{\mathrm{d}_1}(v) - \theta_{\mathrm{d}_0}(v) = \mathbb{E}\{Y^{\mathrm{d}_1} - Y^{\mathrm{d}_0} \mid V = v\}.$$

When V is discrete with support $\mathcal{V}$, the target is the finite collection

$$\{\theta_{\mathrm{d}}(v) : v \in \mathcal{V}\}$$

or, for a contrast of two policies,

$$\{\tau(v) : v \in \mathcal{V}\}.$$

In this case, the target is a finite-dimensional parameter, and standard semiparametric efficiency theory can be used to construct asymptotically linear estimators and Wald-type confidence intervals.

When V is continuous, the target $v \mapsto \theta_{\mathrm{d}}(v)$ or $v \mapsto \tau(v)$ is an infinite-dimensional conditional mean function. In this case, the conditional effect itself is not generally a pathwise differentiable parameter at a point v without additional smoothing or projection. We therefore distinguish between finite-dimensional subgroup inference and flexible estimation of conditional effect functions.

2.4 Identification assumptions

Identification of the conditional LMTP mean relies on a longitudinal causal model, a support condition for the modified treatment policy, and regularity conditions ensuring that the post-policy treatment density is well-defined.

We first introduce a nonparametric structural equation model to state the sequential randomization condition precisely. Suppose the observed data are generated by structural equations

$$L_0 = f_{L_0}(U_{L_0}),$$

and, for $t = 0, \ldots, \tau$,

$$A_t = f_{A_t}(H_t, U_{A_t}), \qquad L_{t+1} = f_{L_{t+1}}(H_t, A_t, U_{L_{t+1}}),$$

where $L_{\tau+1} = Y$. For $t = 0, \ldots, \tau$, define

$$\underline{U}_{L,t+1} = (U_{L_{t+1}}, \ldots, U_{L_{\tau+1}}), \qquad \underline{U}_{A,t+1} = (U_{A_{t+1}}, \ldots, U_{A_\tau}).$$

The exogenous variables encode sources of variation not included in the observed history.

A1 (Consistency)**.** The observed data are generated by the structural equations above. For any treatment history $\overline{a}_\tau$, the counterfactual outcome $Y(\overline{a}_\tau)$ is obtained by replacing the treatment equations with $A_t = a_t$, $t = 0, \ldots, \tau$, while leaving the covariate and outcome equations unchanged. In particular, for the observed treatment history,

$$Y = Y(\overline{A}_\tau).$$

A2 (Strong sequential randomization)**.** For each $t = 0, \ldots, \tau$,

$$U_{A_t} \perp\!\!\!\perp \left(\underline{U}_{L,t+1}, \underline{U}_{A,t+1}\right) \mid H_t.$$

Equivalently, conditional on the observed history H_t, the unobserved determinants of treatment at time t are independent of the unobserved determinants of future covariates, future treatments, and the final outcome.

Remark 1. *Assumption A2 is the NPSEM version of no unmeasured time-varying confounding. It implies the more familiar sequential exchangeability statement that, conditional on H_t, the treatment actually received at time t is independent of future counterfactual variables under the treatment regimes considered in the analysis.*

A3 (LMTP support condition)**.** For each $t = 0, \ldots, \tau$, if

$$(a_t, h_t) \in \mathrm{supp}(A_t, H_t),$$

then

$$\{d_t(a_t, h_t), h_t\} \in \mathrm{supp}(A_t, H_t).$$

Equivalently, for P-almost every history h_t, the modified treatment value $d_t(a_t, h_t)$ lies in the conditional support of $A_t \mid H_t = h_t$ whenever a_t lies in that support.

Remark 2. *Assumption A3 is the LMTP analogue of positivity. It is weaker and more local than the usual static-treatment positivity requirement. It does not require every treatment level to be possible at every history. Instead, it requires only that the treatment value assigned by the modified policy be observable, in principle, among units with the same history.*

A4 (Regularity of the policy map)**.** For each $t = 0, \ldots, \tau$, the map $d_t(\cdot, h_t)$ is known. If A_t is continuous, then $d_t(\cdot, h_t)$ is piecewise smooth and invertible on a finite partition of the treatment space for P-almost every h_t. On each piece, the inverse map $d_t^{-1}(\cdot, h_t)$ exists and has derivative bounded away from zero and infinity on the relevant support. If A_t is discrete, the pushforward law induced by d_t is well-defined.

Under Assumption A4, the post-policy treatment density is defined as the law of $A_t^{\mathrm{d}} = d_t(A_t, H_t)$ conditional on H_t. For continuous treatments, this density is obtained by the change-of-variables formula. In the one-to-one case,

$$g_t^{\mathrm{d}}(a_t \mid h_t) = g_t\{d_t^{-1}(a_t, h_t) \mid h_t\} \left| \frac{\partial}{\partial a_t} d_t^{-1}(a_t, h_t) \right|.$$

For piecewise invertible maps, the density is the sum of the corresponding piecewise inverse contributions. For discrete treatments,

$$g_t^{\mathrm{d}}(a_t \mid h_t) = \sum_{a_t' :\, d_t(a_t', h_t) = a_t} g_t(a_t' \mid h_t).$$

A5 (Baseline conditioning event)**.** For discrete V, assume

$$p_v = \mathbb{P}(V = v) > 0$$

for every subgroup value v under consideration. For continuous V, $\theta_{\mathrm{d}}(v)$ is interpreted as a version of $\mathbb{E}_{P^{\mathrm{d}}}(Y \mid V = v)$, defined for v in the support of V.

Assumptions A1 and A2 give the causal interpretation of the post-policy law. Assumption A3 ensures that the policy does not assign treatment values outside the observed conditional support. Assumption A4 ensures that the post-policy treatment law g_t^{d} and density ratios used for estimation are well-defined. Assumption A5 ensures that the conditional target is defined for the subgroup or covariate values of interest.

2.5 Iterated conditional mean representation

We extend the identification result described in Díaz et al. (????) to the conditional effects of LMTPs now. We carry out a backwards sequential iterated expectations argument, and hence start with the last of a series of regressions. Define the final outcome regression

$$m_\tau^{\mathrm{d}}(a_\tau, h_\tau) = \mathbb{E}_P\left[Y \mid A_\tau = a_\tau, H_\tau = h_\tau\right].$$

For $t = \tau - 1, \ldots, 0$, define the sequential regressions recursively by

$$m_t^{\mathrm{d}}(a_t, h_t) = \mathbb{E}_P\left[m_{t+1}^{\mathrm{d}}\{A_{t+1}^{\mathrm{d}}, H_{t+1}\} \,\middle|\, A_t = a_t, H_t = h_t\right].$$

Here

$$A_{t+1}^{\mathrm{d}} = d_{t+1}(A_{t+1}, H_{t+1})$$

is the treatment value assigned by the modified treatment policy at time $t+1$, evaluated using the observed history H_{t+1}. This recursion is the iterated conditional mean version of the longitudinal g-formula.

With this notation, the marginal LMTP mean is identified as

$$\psi_{\mathrm{d}} = \mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0)\right],$$

and the conditional LMTP mean is identified as

$$\theta_{\mathrm{d}}(v) = \mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) \mid V = v\right].$$

Theorem 1 (Iterated conditional mean identification)**.** *Under the assumptions A1–A5,*

$$\theta_{\mathrm{d}}(v) = \mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) \mid V = v\right]. \tag{1}$$

For two policies d_1 *and* d_0*,*

$$\tau(v) = \mathbb{E}_P\left[m_0^{\mathrm{d}_1}(A_0^{\mathrm{d}_1}, L_0) - m_0^{\mathrm{d}_0}(A_0^{\mathrm{d}_0}, L_0) \mid V = v\right].$$

Proof. The result follows by repeated application of the law of iterated expectations. At the final time point,

$$m_\tau^{\mathrm{d}}(a_\tau, h_\tau) = \mathbb{E}_P(Y \mid A_\tau = a_\tau, H_\tau = h_\tau)$$

is the conditional mean of the outcome given the final treatment and history. At time $\tau - 1$, the regression

$$m_{\tau-1}^{\mathrm{d}}(a_{\tau-1}, h_{\tau-1}) = \mathbb{E}_P\left[m_\tau^{\mathrm{d}}(A_\tau^{\mathrm{d}}, H_\tau) \mid A_{\tau-1} = a_{\tau-1}, H_{\tau-1} = h_{\tau-1}\right]$$

integrates the final outcome regression over the conditional law of the next covariate and treatment, with the next treatment evaluated under the modified policy. Repeating this argument backward through $t = \tau - 2, \ldots, 0$ gives the g-formula mean under P^{d}. Conditioning on $V = v$ gives 1. The contrast result follows by applying the same argument to each policy and taking the difference. □

2.6 Uncentered influence-function pseudo-outcomes

The preceding representations identify the conditional LMTP mean. For estimation, it is useful to construct a pseudo-outcome whose conditional expectation is the target function $v \mapsto \theta_{\mathrm{d}}(v)$. This pseudo-outcome is obtained from the efficient influence function for the marginal LMTP mean by adding back the marginal mean.

Let

$$\eta^{\mathrm{d}} = \left(m_0^{\mathrm{d}}, \ldots, m_\tau^{\mathrm{d}}, r_0^{\mathrm{d}}, \ldots, r_\tau^{\mathrm{d}}\right)$$

denote the nuisance functions for policy d. Define

$$\omega_s^{\mathrm{d}}(O) = \prod_{k=0}^{s} r_k^{\mathrm{d}}(A_k, H_k), \qquad s = 0, \ldots, \tau.$$

For $s = 0, \ldots, \tau - 1$, define the sequential residual

$$\Delta_s^{\mathrm{d}}(O; \eta^{\mathrm{d}}) = m_{s+1}^{\mathrm{d}}\{A_{s+1}^{\mathrm{d}}, H_{s+1}\} - m_s^{\mathrm{d}}(A_s, H_s),$$

and define the terminal residual

$$\Delta_\tau^{\mathrm{d}}(O; \eta^{\mathrm{d}}) = Y - m_\tau^{\mathrm{d}}(A_\tau, H_\tau).$$

Thus each residual compares the next-stage pseudo-outcome to its conditional mean given the current treatment and history.

The uncentered influence-function pseudo-outcome is

$$\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) = m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) + \sum_{s=0}^{\tau} \omega_s^{\mathrm{d}}(O) \Delta_s^{\mathrm{d}}(O; \eta^{\mathrm{d}}).$$

Equivalently,

$$\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) = m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) + \sum_{s=0}^{\tau-1} \left\{ \prod_{k=0}^{s} r_k^{\mathrm{d}}(A_k, H_k) \right\} \left[m_{s+1}^{\mathrm{d}}\{A_{s+1}^{\mathrm{d}}, H_{s+1}\} - m_s^{\mathrm{d}}(A_s, H_s) \right] + \left\{ \prod_{k=0}^{\tau} r_k^{\mathrm{d}}(A_k, H_k) \right\} \left[Y - m_\tau^{\mathrm{d}}(A_\tau, H_\tau) \right].$$

The product in this expression is the product of density ratios up to time s; the terms in square brackets are sequential regression residuals. Thus the EIF is a sum of weighted residuals.

At the true nuisance functions, the centered efficient influence function for the marginal LMTP mean ψ_{d} is

$$D_{\mathrm{d}}(O) = \varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) - \psi_{\mathrm{d}}.$$

Equivalently,

$$\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) = \psi_{\mathrm{d}} + D_{\mathrm{d}}(O).$$

Therefore

$$\mathbb{E}_P\{\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}})\} = \psi_{\mathrm{d}}.$$

The key property for conditional effect estimation is that, since $V \subseteq L_0$, the conditional mean of the uncentered pseudo-outcome is the conditional LMTP mean.

Theorem 2 (Pseudo-outcome representation of the conditional LMTP mean)**.** *Under Assumptions A1–A5,*

$$\mathbb{E}_P\left[\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) \mid V = v\right] = \theta_{\mathrm{d}}(v).$$

Therefore, for two policies d_1 *and* d_0*, if*

$$\varphi_\tau(O; \eta) = \varphi_{\mathrm{d}_1}(O; \eta^{\mathrm{d}_1}) - \varphi_{\mathrm{d}_0}(O; \eta^{\mathrm{d}_0}),$$

then

$$\mathbb{E}_P\left[\varphi_\tau(O; \eta) \mid V = v\right] = \tau(v).$$

Proof. By Theorem 1,

$$\mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) \mid V = v\right] = \theta_{\mathrm{d}}(v).$$

It remains to show that the weighted residual terms have conditional mean zero given V.

Fix $s \in \{0, \ldots, \tau - 1\}$. Since $\omega_s^{\mathrm{d}}(O)$ is measurable with respect to (A_s, H_s), and since $V \subseteq L_0 \subseteq H_s$, iterated expectation gives

$$\mathbb{E}_P\left[\omega_s^{\mathrm{d}}(O)\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V\right] = \mathbb{E}_P\left[\omega_s^{\mathrm{d}}(O)\mathbb{E}_P\left\{\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid A_s, H_s\right\} \mid V\right].$$

By the definition of m_s^{d},

$$\mathbb{E}_P\left[m_{s+1}^{\mathrm{d}}\{A_{s+1}^{\mathrm{d}}, H_{s+1}\} \mid A_s, H_s\right] = m_s^{\mathrm{d}}(A_s, H_s),$$

so

$$\mathbb{E}_P\left\{\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid A_s, H_s\right\} = 0.$$

Therefore

$$\mathbb{E}_P\left[\omega_s^{\mathrm{d}}(O)\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V\right] = 0.$$

For the terminal term,

$$\mathbb{E}_P\left[Y - m_\tau^{\mathrm{d}}(A_\tau, H_\tau) \mid A_\tau, H_\tau\right] = 0$$

by the definition of m_τ^{d}. The same iterated-expectation argument gives

$$\mathbb{E}_P\left[\omega_\tau^{\mathrm{d}}(O)\Delta_\tau^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V\right] = 0.$$

Thus all weighted residual terms in $\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}})$ have conditional mean zero given V. Hence

$$\mathbb{E}_P\left[\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V = v\right] = \mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) \mid V = v\right] = \theta_{\mathrm{d}}(v).$$

The contrast result follows by linearity of conditional expectation. □

Remark 3 (Role of the pseudo-outcome)**.** *The pseudo-outcome $\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}})$ is not a new causal parameter. It is a representation of the same identified LMTP mean. Its role is statistical: it converts estimation of the conditional LMTP mean into a regression problem. For a general conditioning variable V, the DR-Learner estimates*

$$v \mapsto \mathbb{E}\left[\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V = v\right]$$

by replacing the unknown nuisance functions with estimates and then regressing the resulting estimated pseudo-outcomes on V.

3 Estimation

We now describe estimators for the conditional LMTP parameters introduced in Section 2. We consider two complementary estimation strategies. First, when V is discrete with finite support $\mathcal{V}$, we construct a targeted minimum loss-based estimator of the finite-dimensional vector

$$\{\theta_{\mathrm{d}}(v) : v \in \mathcal{V}\}.$$

This estimator targets the efficient influence-function estimating equations for all subgroups jointly. Second, when V is continuous, high-dimensional, or when the goal is to estimate a smooth conditional effect function, we construct a DR-Learner by regressing estimated uncentered influence-function pseudo-outcomes on V.

Throughout this section, let

$$\eta^{\mathrm{d}} = \left(m_0^{\mathrm{d}}, \ldots, m_\tau^{\mathrm{d}}, r_0^{\mathrm{d}}, \ldots, r_\tau^{\mathrm{d}}\right)$$

denote the nuisance functions for policy d. For two policies d_1 and d_0, all steps below may be applied separately to each policy and then contrasted.

3.1 Cross-fitting and nuisance estimation

Flexible estimation of the nuisance functions is essential in longitudinal settings, where the treatment mechanism and sequential outcome regressions may depend on high-dimensional histories. We use cross-fitting to separate nuisance estimation from evaluation of the estimating equations. This allows the nuisance functions to be estimated with flexible regression or machine-learning methods without requiring empirical process restrictions such as Donsker conditions.

Let

$$\mathcal{I}_1, \ldots, \mathcal{I}_J$$

be a partition of $\{1, \ldots, n\}$ into J folds. For each fold j, let $\mathcal{I}_{-j}$ denote its complement. Nuisance functions are fit using only the training sample $\mathcal{I}_{-j}$, and predictions are evaluated only on the validation sample $\mathcal{I}_j$. For observation i, let $j(i)$ denote the fold containing i.

For each policy $\mathbb{d}$, each fold j, and each time point $t = 0, \ldots, \tau$, estimate the density ratio

$$r_t^{\mathbb{d}}(a_t, h_t) = \frac{g_t^{\mathbb{d}}(a_t \mid h_t)}{g_t(a_t \mid h_t)}$$

using the training sample $\mathcal{I}_{-j}$. Denote the resulting cross-fitted prediction for observation i by

$$\widehat{r}_{t,i}^{\mathbb{d}} = \widehat{r}_{t,j(i)}^{\mathbb{d}}(A_{t,i}, H_{t,i}).$$

The estimated cumulative density ratio through time s is

$$\widehat{\omega}_{s,i}^{\mathbb{d}} = \prod_{t=0}^{s} \widehat{r}_{t,i}^{\mathbb{d}}.$$

The sequential outcome regressions are also estimated recursively using cross-fitting. Starting at the final time point, estimate

$$m_\tau^{\mathbb{d}}(a_\tau, h_\tau) = \mathbb{E}(Y \mid A_\tau = a_\tau, H_\tau = h_\tau)$$

on $\mathcal{I}_{-j}$, and evaluate it on $\mathcal{I}_j$ at both the observed and policy-modified treatment values as required. For $t = \tau - 1, \ldots, 0$, estimate

$$m_t^{\mathbb{d}}(a_t, h_t) = \mathbb{E}\left[m_{t+1}^{\mathbb{d}}\{A_{t+1}^{\mathbb{d}}, H_{t+1}\} \mid A_t = a_t, H_t = h_t\right].$$

In practice, the regression outcome at time t is the previously estimated next-stage pseudo-outcome

$$\widehat{m}_{t+1,j}^{\mathbb{d}}\{A_{t+1}^{\mathbb{d}}, H_{t+1}\},$$

constructed using the training sample. The resulting cross-fitted predictions for observation i are denoted by

$$\widehat{m}_{t,i}^{\mathbb{d}}(A_{t,i}, H_{t,i}) \qquad \text{and} \qquad \widehat{m}_{t,i}^{\mathbb{d}}(A_{t,i}^{\mathbb{d}}, H_{t,i}).$$

Given these nuisance estimates, define the cross-fitted estimated residuals

$$\widehat{\Delta}_{s,i}^{\mathbb{d}} = \widehat{m}_{s+1,i}^{\mathbb{d}}\{A_{s+1,i}^{\mathbb{d}}, H_{s+1,i}\} - \widehat{m}_{s,i}^{\mathbb{d}}(A_{s,i}, H_{s,i}), \qquad s = 0, \ldots, \tau - 1,$$

and

$$\widehat{\Delta}_{\tau,i}^{\mathbb{d}} = Y_i - \widehat{m}_{\tau,i}^{\mathbb{d}}(A_{\tau,i}, H_{\tau,i}).$$

The estimated uncentered influence-function pseudo-outcome is then

$$\widehat{\varphi}_{\mathbb{d},i} = \widehat{m}_{0,i}^{\mathbb{d}}(A_{0,i}^{\mathbb{d}}, L_{0,i}) + \sum_{s=0}^{\tau} \widehat{\omega}_{s,i}^{\mathbb{d}} \widehat{\Delta}_{s,i}^{\mathbb{d}}.$$

For a contrast of two policies, define

$$\widehat{\varphi}_{\tau,i} = \widehat{\varphi}_{\mathbb{d}_1,i} - \widehat{\varphi}_{\mathbb{d}_0,i}.$$

Remark 4. *The notation $\widehat{\varphi}_{\tau,i}$ denotes the estimated pseudo-outcome for the conditional contrast $\tau(v)$, not a time-indexed quantity. When there is risk of confusion, we write $\widehat{\varphi}_{\mathbb{d}_1-\mathbb{d}_0,i}$.*

3.2 Targeted estimation for finite baseline subgroups

Suppose V is discrete with finite support

$$\mathcal{V} = \{v_1, \ldots, v_K\},$$

and let

$$p_v = \mathbb{P}(V = v).$$

The conditional LMTP mean for subgroup v can be written as

$$\theta_{\mathrm{d}}(v) = \frac{\mathbb{E}\left[\mathbb{1}(V = v) m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0)\right]}{p_v}.$$

The efficient influence function for $\theta_{\mathrm{d}}(v)$ is

$$D_{\mathrm{d},v}(O) = \frac{\mathbb{1}(V = v)}{p_v} \left\{\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) - \theta_{\mathrm{d}}(v)\right\}.$$

Equivalently,

$$D_{\mathrm{d},v}(O) = \frac{\mathbb{1}(V = v)}{p_v} \left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) - \theta_{\mathrm{d}}(v) + \sum_{s=0}^{\tau} \omega_s^{\mathrm{d}}(O) \Delta_s^{\mathrm{d}}(O; \eta^{\mathrm{d}})\right].$$

For a contrast

$$\tau(v) = \theta_{\mathrm{d}_1}(v) - \theta_{\mathrm{d}_0}(v),$$

the efficient influence function is

$$D_{\tau,v}(O) = D_{\mathrm{d}_1,v}(O) - D_{\mathrm{d}_0,v}(O).$$

The subgroup TMLE targets the sequential outcome regressions so that the empirical mean of the estimated efficient influence function is approximately zero for each $v \in \mathcal{V}$. Let

$$\widehat{p}_v = \mathbb{P}_n\{\mathbb{1}(V = v)\}.$$

For $s = 0, \ldots, \tau$, define the subgroup-specific clever covariate

$$\widehat{H}_{s,v}^{\mathrm{d}}(O_i) = \frac{\mathbb{1}(V_i = v)}{\widehat{p}_v} \widehat{\omega}_{s,i}^{\mathrm{d}}.$$

Collecting these across $v \in \mathcal{V}$ gives the vector clever covariate

$$\widehat{H}_s^{\mathrm{d}}(O_i) = \left(\widehat{H}_{s,v_1}^{\mathrm{d}}(O_i), \ldots, \widehat{H}_{s,v_K}^{\mathrm{d}}(O_i)\right)^\top.$$

The targeting step proceeds backward in time. Let

$$\widehat{Y}_{\tau+1,i}^{\mathrm{d},*} = Y_i.$$

For $s = \tau, \tau - 1, \ldots, 0$, fit a fluctuation submodel for the current outcome regression using $\widehat{Y}_{s+1,i}^{\mathrm{d},*}$ as the pseudo-outcome, $\widehat{m}_{s,i}^{\mathrm{d}}(A_{s,i}, H_{s,i})$ as the initial prediction, and $\widehat{H}_s^{\mathrm{d}}(O_i)$ as the clever covariate. With an identity link, the fluctuation model is

$$\widehat{m}_{s,\epsilon}^{\mathrm{d}}(A_{s,i}, H_{s,i}) = \widehat{m}_s^{\mathrm{d}}(A_{s,i}, H_{s,i}) + \epsilon_s^\top \widehat{H}_s^{\mathrm{d}}(O_i),$$

where $\epsilon_s \in \mathbb{R}^K$. With a bounded outcome and logit link, the fluctuation model is

$$\operatorname{logit} \widehat{m}_{s,\epsilon}^{\mathrm{d}}(A_{s,i}, H_{s,i}) = \operatorname{logit} \widehat{m}_s^{\mathrm{d}}(A_{s,i}, H_{s,i}) + \epsilon_s^\top \widehat{H}_s^{\mathrm{d}}(O_i).$$

Let $\widehat{\epsilon}_s$ denote the fitted fluctuation coefficient and write

$$\widehat{m}_s^{\mathrm{d},*} = \widehat{m}_{s,\widehat{\epsilon}_s}^{\mathrm{d}}.$$

After updating the regression at time s, define the next outcome for the backward recursion by evaluating the updated regression under the modified treatment value:

$$\widehat{Y}_{s,i}^{\mathrm{d},*} = \widehat{m}_s^{\mathrm{d},*}(A_{s,i}^{\mathrm{d}}, H_{s,i}).$$

This updated pseudo-outcome is then used in the fluctuation at time $s - 1$.

The subgroup TMLE of the policy-specific conditional mean is the plug-in estimator

$$\widehat{\theta}_{\mathrm{d},\mathrm{tmle}}(v) = \mathbb{P}_n \left[\frac{\mathbb{1}(V = v)}{\widehat{p}_v} \widehat{m}_0^{\mathrm{d},*}(A_0^{\mathrm{d}}, L_0) \right].$$

For two policies d_1 and d_0, the contrast estimator is

$$\widehat{\tau}_{\mathrm{tmle}}(v) = \widehat{\theta}_{\mathrm{d}_1,\mathrm{tmle}}(v) - \widehat{\theta}_{\mathrm{d}_0,\mathrm{tmle}}(v).$$

The estimated efficient influence function for the policy-specific subgroup mean is

$$\widehat{D}_{\mathrm{d},v}(O_i) = \frac{\mathbb{1}(V_i = v)}{\widehat{p}_v} \left[\widehat{m}_0^{\mathrm{d},*}(A_{0,i}^{\mathrm{d}}, L_{0,i}) - \widehat{\theta}_{\mathrm{d},\mathrm{tmle}}(v) + \sum_{s=0}^{\tau} \widehat{\omega}_{s,i}^{\mathrm{d}} \widehat{\Delta}_{s,i}^{\mathrm{d},*} \right],$$

where

$$\widehat{\Delta}_{s,i}^{\mathrm{d},*} = \widehat{m}_{s+1}^{\mathrm{d},*}(A_{s+1,i}^{\mathrm{d}}, H_{s+1,i}) - \widehat{m}_s^{\mathrm{d},*}(A_{s,i}, H_{s,i}), \qquad s = 0, \ldots, \tau - 1,$$

and

$$\widehat{\Delta}_{\tau,i}^{\mathrm{d},*} = Y_i - \widehat{m}_{\tau}^{\mathrm{d},*}(A_{\tau,i}, H_{\tau,i}).$$

For a contrast, define

$$\widehat{D}_{\tau,v}(O_i) = \widehat{D}_{\mathrm{d}_1,v}(O_i) - \widehat{D}_{\mathrm{d}_0,v}(O_i).$$

For a single subgroup v, the estimated standard error of $\widehat{\tau}_{\mathrm{tmle}}(v)$ is

$$\widehat{\mathrm{se}}\{\widehat{\tau}_{\mathrm{tmle}}(v)\} = \left[\frac{1}{n} \mathbb{P}_n \left\{ \widehat{D}_{\tau,v}(O)^2 \right\} \right]^{1/2}.$$

For simultaneous inference over $v \in \mathcal{V}$, define the empirical covariance matrix

$$\widehat{\Sigma}_{\tau} = \mathbb{P}_n \left[\widehat{D}_{\tau}(O) \widehat{D}_{\tau}(O)^{\top} \right],$$

where

$$\widehat{D}_{\tau}(O) = \left(\widehat{D}_{\tau,v_1}(O), \ldots, \widehat{D}_{\tau,v_K}(O) \right)^{\top}.$$

Then

$$\sqrt{n} \left\{ \widehat{\tau}_{\mathrm{tmle}} - \tau \right\}$$

is approximated by a mean-zero multivariate normal distribution with covariance matrix $\widehat{\Sigma}_{\tau}$. Simultaneous confidence intervals may be formed using the quantiles of

$$\max_{v \in \mathcal{V}} \left| Z_v / \widehat{\sigma}_v \right|, \qquad Z \sim N(0, \widehat{\Sigma}_{\tau}),$$

where $\widehat{\sigma}_v^2$ is the vth diagonal element of $\widehat{\Sigma}_{\tau}$.

3.3 DR-Learner for conditional LMTP effects

When V is continuous or high-dimensional, the target

$$v \mapsto \theta_{\mathrm{d}}(v)$$

is a conditional mean function rather than a finite-dimensional parameter. The pseudo-outcome representation in Theorem 2 suggests a direct meta-learning approach: estimate the uncentered influence-function pseudo-outcome and regress it on V.

For a single policy d, the DR-Learner proceeds as follows.

1. Split the sample into folds $\mathcal{I}_1, \ldots, \mathcal{I}_J$.
2. On each training sample $\mathcal{I}_{-j}$, estimate the nuisance functions

$$\eta^{\mathrm{d}} = (m_0^{\mathrm{d}}, \ldots, m_{\tau}^{\mathrm{d}}, r_0^{\mathrm{d}}, \ldots, r_{\tau}^{\mathrm{d}}).$$

3. For each observation $i \in \mathcal{I}_j$, construct the cross-fitted pseudo-outcome

$$\widehat{\varphi}_{\mathrm{d},i} = \widehat{m}^{\mathrm{d}}_{0,i}(A^{\mathrm{d}}_{0,i}, L_{0,i}) + \sum_{s=0}^{\tau} \widehat{\omega}^{\mathrm{d}}_{s,i} \widehat{\Delta}^{\mathrm{d}}_{s,i}.$$

4. Regress $\widehat{\varphi}_{\mathrm{d},i}$ on V_i using a chosen second-stage regression method. The fitted regression function is

$$\widehat{\theta}_{\mathrm{d},\mathrm{dr}}(v) = \widehat{\mathbb{E}}_n \left[\widehat{\varphi}_{\mathrm{d}} \mid V = v \right].$$

For two policies d_1 and d_0, construct the contrast pseudo-outcome

$$\widehat{\varphi}_{\tau,i} = \widehat{\varphi}_{\mathrm{d}_1,i} - \widehat{\varphi}_{\mathrm{d}_0,i},$$

and regress it on V_i:

$$\widehat{\tau}_{\mathrm{dr}}(v) = \widehat{\mathbb{E}}_n \left[\widehat{\varphi}_{\tau} \mid V = v \right].$$

Equivalently,

$$\widehat{\tau}_{\mathrm{dr}}(v) = \widehat{\theta}_{\mathrm{d}_1,\mathrm{dr}}(v) - \widehat{\theta}_{\mathrm{d}_0,\mathrm{dr}}(v),$$

if the same second-stage regression method is applied separately to the two policy-specific pseudo-outcomes.

The second-stage regression may be chosen to reflect the scientific structure expected in the conditional effect. If V is low-dimensional and continuous, local linear regression, splines, generalized additive models, or local polynomial regression are natural choices. We note that the second stage learners should satisfy the linear smoother definition given in Kennedy (????).

The corresponding benchmark estimator is the regression that would be obtained if the nuisance functions were known:

$$\widetilde{\theta}_{\mathrm{d}}(v) = \widehat{\mathbb{E}}_n \left[\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) \mid V = v \right],$$

and, for a contrast,

$$\widetilde{\tau}(v) = \widehat{\mathbb{E}}_n \left[\varphi_{\tau}(O; \eta) \mid V = v \right].$$

The theoretical results in Section 4 compare the feasible DR-Learner to this benchmark regression. In particular, oracle equivalence holds when the smoothed second-order bias induced by nuisance estimation is negligible relative to the oracle second-stage regression error.

Remark 5. *For finite discrete V, the DR-Learner with a saturated second-stage regression is simply the subgroup mean of the estimated pseudo-outcomes. Thus the finite-subgroup estimator can be viewed as the discrete special case of the pseudo-outcome regression strategy. The targeted estimator, however, has the additional advantage that it updates the nuisance regressions to solve the efficient influence-function equations for the subgroup parameters.*

3.4 Projections onto working marginal structural models

The subgroup TMLE of Section 3.2 and the DR-Learner of Section 3.3 reflect two ends of a spectrum: on the one hand, fully saturated inference over a finite set of discrete subgroups $\mathcal{V}$, and on the other fully nonparametric estimation of $v \mapsto \theta_{\mathrm{d}}(v)$ at nonparametric rates. Applied analysis often calls for something in between, such as a low-dimensional, interpretable summary of how the policy effect varies with V, with root-n inference. We note that both of these procedures are obtained by regressing the efficient influence function pseudo-outcome on a basis in V: a fixed basis yields a working marginal structural model (MSM) whose projection coefficients are pathwise differentiable, while a basis of growing dimension, undersmoothed so that approximation bias is negligible, can yield valid inference on θ_{d} itself. We explore the idea of projecting onto a working MSM here.

3.5 Fixed-dimensional working models

Let $b(v) = \{b_1(v), \ldots, b_k(v)\}^\top$ be a fixed, user-specified vector of basis functions — for example $b(v) = (1, v)^\top$ for a linear trend in a scalar modifier, or a coarse spline basis — and let $h(v) \geq 0$ be a user-specified weight function, with $h \equiv 1$ as the default. Define the working-model projection of the conditional LMTP mean,

$$\beta_{\mathrm{d}} = \arg\min_{\beta \in \mathbb{R}^k} \mathbb{E}\left[h(V) \left\{ \theta_{\mathrm{d}}(V) - b(V)^\top \beta \right\}^2 \right] = M^{-1} \mathbb{E}\left\{ h(V)\, b(V)\, \theta_{\mathrm{d}}(V) \right\}, \qquad M = \mathbb{E}\left\{ h(V)\, b(V)\, b(V)^\top \right\},$$

assuming M is nonsingular. The parameter β_{d} is defined without assuming the working model is correct: $v \mapsto b(v)^\top \beta_{\mathrm{d}}$ is the weighted least-squares projection of θ_{d} onto the span of b, and retains this interpretation under arbitrary misspecification **??**. When the working model happens to hold, $b(v)^\top \beta_{\mathrm{d}} = \theta_{\mathrm{d}}(v)$. For a contrast of two policies,

$\beta_\tau = \beta_{\mathrm{d}_1} - \beta_{\mathrm{d}_0}$ is the projection of $\tau(\cdot)$, obtained by replacing θ_{d} with τ throughout, or equivalently by using the contrast pseudo-outcome $\varphi_\tau = \varphi_{\mathrm{d}_1} - \varphi_{\mathrm{d}_0}$.

Because $\mathbb{E}\{\varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}}) \mid V = v\} = \theta_{\mathrm{d}}(v)$ by Theorem 2, the projection coefficients admit the moment representation $\beta_{\mathrm{d}} = M^{-1}\mathbb{E}\{h(V)b(V)\varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}})\}$, which is a smooth finite-dimensional functional of the observed data law.

Proposition 1 (Efficient influence function of the projection)**.** *Under Assumptions A1–A5 and nonsingular M, the parameter β_{d} is pathwise differentiable with efficient influence function*

$$D_{\beta,\mathrm{d}}(O) = M^{-1}\, h(V)\, b(V)\, \left\{\varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}}) - b(V)^\top \beta_{\mathrm{d}}\right\}.$$

For the contrast, $D_{\beta,\tau} = D_{\beta,\mathrm{d}_1} - D_{\beta,\mathrm{d}_0}$.

Proof. The parameter solves the moment equation $\mathbb{E}\, U(O;\beta) = 0$ with $U(O;\beta) = h(V)b(V)\{\varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}}) - b(V)^\top\beta\}$. The functional $P \mapsto \mathbb{E}_P\{h(V)b(V)\theta_{\mathrm{d},P}(V)\}$ has efficient influence function $h(V)b(V)\varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}})$ minus its mean, since φ_{d} is the uncentered efficient influence function for the LMTP mean and $h\,b$ is a fixed baseline-measurable weight; the contribution of the marginal law of V through M is absorbed by centering at $b(V)^\top\beta_{\mathrm{d}}$, which renders $D_{\beta,\mathrm{d}}$ mean zero by the normal equations. The delta method for estimating equations, with derivative matrix $\partial_\beta\, \mathbb{E}\, U = -M$, gives the stated form. □

The estimator is the weighted least-squares regression of the cross-fitted pseudo-outcomes of Section 3.1 on the basis:

$$\widehat{\beta}_{\mathrm{d}} = \widehat{M}^{-1}\, \mathbb{P}_n\, \{h(V)\, b(V)\, \widehat{\varphi}_{\mathrm{d}}(O)\}, \qquad \widehat{M} = \mathbb{P}_n\left\{h(V)\, b(V)\, b(V)^\top\right\}.$$

Theorem 3 (Asymptotic linearity of the working-model projection)**.** *Suppose Assumptions A1–A5 hold, M is nonsingular, h and b are bounded, the pseudo-outcomes are constructed with cross-fitting, the cumulative estimated ratios are bounded in probability, and the nuisance product-rate condition*

$$\sum_{s=0}^{\tau} \left\|\widehat{r}_s^{\mathrm{d}} - r_s^{\mathrm{d}}\right\|_2 \left\|\widehat{m}_s^{\mathrm{d}} - m_s^{\mathrm{d}}\right\|_2 = o_p(n^{-1/2})$$

holds in the marginal $L_2(P_0)$ norm. Then

$$\widehat{\beta}_{\mathrm{d}} - \beta_{\mathrm{d}} = \mathbb{P}_n D_{\beta,\mathrm{d}} + o_p(n^{-1/2}), \qquad \sqrt{n}\left(\widehat{\beta}_{\mathrm{d}} - \beta_{\mathrm{d}}\right) \rightsquigarrow N(0,\Sigma_\beta), \qquad \Sigma_\beta = \mathbb{E}\left\{D_{\beta,\mathrm{d}} D_{\beta,\mathrm{d}}^\top\right\},$$

and Σ_β is consistently estimated by the sandwich estimator

$$\widehat{\Sigma}_\beta = \widehat{M}^{-1}\, \mathbb{P}_n\left[h(V)^2\, b(V)\, b(V)^\top \left\{\widehat{\varphi}_{\mathrm{d}}(O) - b(V)^\top\widehat{\beta}_{\mathrm{d}}\right\}^2\right] \widehat{M}^{-1}.$$

The same statements hold for the contrast with φ_τ, $D_{\beta,\tau}$, and β_τ.

Proof. Writing $\widehat{\beta}_{\mathrm{d}} - \beta_{\mathrm{d}} = \widehat{M}^{-1}\mathbb{P}_n\, h\, b\, \{\widehat{\varphi}_{\mathrm{d}} - b^\top\beta_{\mathrm{d}}\}$ and decomposing $\widehat{\varphi}_{\mathrm{d}} = \varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}}) + b_{\widehat{\eta}}^{\mathrm{d}}(V) + e_{\widehat{\eta}}(O)$ as in the proof of Theorem 5, the term involving $\varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}})$ yields the influence-function expansion, the conditionally mean-zero term $e_{\widehat{\eta}}$ is $o_p(n^{-1/2})$ by cross-fitting, and the bias term is bounded by $\|h\,b\|_\infty\, \mathbb{E}|b_{\widehat{\eta}}^{\mathrm{d}}(V)|$, which is $o_p(n^{-1/2})$ under the product-rate condition by Proposition **??** and the Cauchy–Schwarz inequality; $\widehat{M} \to M$ in probability completes the argument. □

Remark 6 (Simultaneous bands for the fitted summary)**.** *Wald intervals for individual coefficients (for example, a linear-trend slope) follow directly. For the fitted curve, a simultaneous band over any finite grid $v_1, \ldots, v_G$ is $b(v_g)^\top\widehat{\beta}_{\mathrm{d}} \pm c_{1-\alpha}\{b(v_g)^\top\widehat{\Sigma}_\beta\, b(v_g)/n\}^{1/2}$, where $c_{1-\alpha}$ is the max-$|Z|$ critical value computed from the Gaussian vector with correlation matrix induced by $\{b(v_g)^\top\widehat{\Sigma}_\beta\, b(v_{g'})\}_{g,g'}$, exactly as in Section 3.2. Homogeneity of the effect in V within the working model corresponds to the null that all non-intercept coordinates of β_τ vanish, testable by a Wald statistic based on $\widehat{\Sigma}_\beta$.*

Remark 7 (The subgroup TMLE as a saturated projection)**.** *When V is discrete with support $\{v_1, \ldots, v_K\}$, taking $h \equiv 1$ and the saturated basis $b(v) = \{\mathbb{1}(v = v_1), \ldots, \mathbb{1}(v = v_K)\}^\top$ gives $\widehat{\beta}_{\mathrm{d},k} = \mathbb{P}_n\{\mathbb{1}(V = v_k)\widehat{\varphi}_{\mathrm{d}}\}/\widehat{p}_{v_k}$, the one-step subgroup estimator, with $D_{\beta,\mathrm{d}}$ reducing coordinatewise to the subgroup efficient influence functions $D_{\mathrm{d},v}$ of Section 3.2. The working-model formulation therefore nests the subgroup analysis, and the TMLE of Section 3.2 may be viewed as its targeted refinement, whose plug-in form guarantees that estimates respect the parameter space.*

4 Theoretical Properties

We now describe the asymptotic properties of the proposed estimators. The finite subgroup TMLE is a regular asymptotically linear estimator of the vector of subgroup-specific LMTP effects under standard cross-fitting and second-order remainder conditions. The DR-Learner is analyzed as a regression with estimated pseudo-outcomes. Its behavior is compared to the benchmark regression that would be obtained if the nuisance functions were known.

Throughout this section, let P_0 denote the true observed data law. For a policy $\mathbb{d}$, let $\eta_0^{\mathbb{d}}$ denote the corresponding true nuisance functions, and let

$$\varphi_{\mathbb{d}}(O; \eta_0^{\mathbb{d}})$$

denote the uncentered LMTP influence-function pseudo-outcome defined in Section 2.6. For a contrast of two policies $\mathbb{d}_1$ and $\mathbb{d}_0$, define

$$\varphi_\tau(O; \eta_0) = \varphi_{\mathbb{d}_1}(O; \eta_0^{\mathbb{d}_1}) - \varphi_{\mathbb{d}_0}(O; \eta_0^{\mathbb{d}_0}).$$

4.1 Asymptotic linearity of the subgroup TMLE

Suppose V is discrete with finite support

$$\mathcal{V} = \{v_1, \ldots, v_K\}.$$

Let

$$\theta_{\mathbb{d}} = \{\theta_{\mathbb{d}}(v_1), \ldots, \theta_{\mathbb{d}}(v_K)\}^\top$$

denote the vector of policy-specific subgroup LMTP means. For a contrast of two policies, write

$$\tau = \{\tau(v_1), \ldots, \tau(v_K)\}^\top.$$

The efficient influence function for the policy-specific subgroup mean $\theta_{\mathbb{d}}(v)$ is

$$D_{\mathbb{d},v}(O) = \frac{\mathbb{1}(V = v)}{p_v} \left\{\varphi_{\mathbb{d}}(O; \eta_0^{\mathbb{d}}) - \theta_{\mathbb{d}}(v)\right\}, \qquad p_v = P_0(V = v).$$

For a contrast,

$$D_{\tau,v}(O) = D_{\mathbb{d}_1,v}(O) - D_{\mathbb{d}_0,v}(O).$$

Let

$$D_\tau(O) = \{D_{\tau,v_1}(O), \ldots, D_{\tau,v_K}(O)\}^\top.$$

A6 (Regularity for subgroup TMLE)**.** The following conditions hold.

1. The identification assumptions in Section 2.4 hold for each policy under consideration.
2. For each $v \in \mathcal{V}$, $p_v > 0$, and $\min_{v \in \mathcal{V}} p_v$ is bounded away from zero.
3. The outcome and sequential regressions are uniformly bounded, and the cumulative density ratios $\omega_s^{\mathbb{d}}(O) = \prod_{t=0}^{s} r_t^{\mathbb{d}}(A_t, H_t)$ are uniformly bounded or have sufficiently many bounded moments.
4. The nuisance functions are estimated using cross-fitting, so that nuisance estimation and evaluation of the estimating equations are performed on independent samples.
5. The targeting step solves the empirical efficient influence-function equations up to negligible error:
$$\mathbb{P}_n \widehat{D}_{\mathbb{d},v} = o_p(n^{-1/2}) \qquad \text{for all } v \in \mathcal{V}.$$
6. The second-order remainder for each policy satisfies
$$R_{\mathbb{d},n}(v) = o_p(n^{-1/2}) \qquad \text{for all } v \in \mathcal{V}.$$

Theorem 4 (Asymptotic linearity of the subgroup TMLE)**.** *Under Assumption A6,*

$$\widehat{\theta}_{\mathbb{d},\text{tmle}}(v) - \theta_{\mathbb{d}}(v) = \mathbb{P}_n D_{\mathbb{d},v} + o_p(n^{-1/2})$$

for each $v \in \mathcal{V}$. Hence

$$\sqrt{n}\left\{\widehat{\theta}_{\mathbb{d},\text{tmle}} - \theta_{\mathbb{d}}\right\} \rightsquigarrow N(0, \Sigma_{\mathbb{d}}),$$

where

$$\Sigma_{\mathbb{d}} = \mathbb{E}_{P_0}\left[D_{\mathbb{d}}(O) D_{\mathbb{d}}(O)^\top\right].$$

For a contrast of two policies,

$$\widehat{\tau}_{\text{tmle}}(v) - \tau(v) = \mathbb{P}_n D_{\tau,v} + o_p(n^{-1/2}),$$

and therefore

$$\sqrt{n}\left\{\widehat{\tau}_{\text{tmle}} - \tau\right\} \rightsquigarrow N(0, \Sigma_\tau),$$

where

$$\Sigma_\tau = \mathbb{E}_{P_0}\left[D_\tau(O) D_\tau(O)^\top\right].$$

Proof. We give the argument for a single policy. The contrast result follows by linearity. By the definition of the subgroup parameter and the efficient influence function,

$$\widehat{\theta}_{\mathbb{d},\text{tmle}}(v) - \theta_{\mathbb{d}}(v) = (\mathbb{P}_n - P_0) D_{\mathbb{d},v} + \mathbb{P}_n \widehat{D}_{\mathbb{d},v} + R_{\mathbb{d},n}(v) + o_p(n^{-1/2}).$$

The empirical estimating equation term $\mathbb{P}_n \widehat{D}_{\mathbb{d},v}$ is $o_p(n^{-1/2})$ by the targeting condition. The second-order remainder is $o_p(n^{-1/2})$ by assumption. The cross-fitting construction removes the need for Donsker restrictions on the estimated nuisance functions and allows the leading empirical process term to be evaluated at the fixed true influence function. Thus

$$\widehat{\theta}_{\mathbb{d},\text{tmle}}(v) - \theta_{\mathbb{d}}(v) = \mathbb{P}_n D_{\mathbb{d},v} + o_p(n^{-1/2}).$$

The multivariate central limit theorem gives the joint normal limit over $v \in \mathcal{V}$. □

Remark 8. *The empirical covariance estimator*

$$\widehat{\Sigma}_\tau = \mathbb{P}_n\left[\widehat{D}_\tau(O)\widehat{D}_\tau(O)^\top\right]$$

is consistent for Σ_τ under the same conditions, provided the estimated influence functions converge in $L_2(P_0)$ to their population limits.

4.2 Oracle equivalence of the DR-Learner

We next study the DR-Learner for the conditional LMTP contrast

$$\tau(v) = \mathbb{E}_{P_0}\{\varphi_\tau(O;\eta_0) \mid V = v\}.$$

Let $\widehat{\varphi}_{\tau,i}$ be the cross-fitted estimated pseudo-outcome defined in Section 3.3. Let

$$\widehat{\mathbb{E}}_n[Z \mid V = v]$$

denote a generic second-stage regression estimator applied to an outcome Z and evaluated at $V = v$. This notation may represent, for example, a linear regression, a series or sieve regression, spline regression, or local polynomial regression. Following Kennedy (????), the second stage regressor needs to be a *linear smoother* (a wide class of methods) for oracle equivalency results to be achieved.

The feasible DR-Learner is

$$\widehat{\tau}_{\text{dr}}(v) = \widehat{\mathbb{E}}_n\left[\widehat{\varphi}_\tau \mid V = v\right].$$

The benchmark regression that would be obtained if the nuisance functions were known is

$$\widetilde{\tau}(v) = \widehat{\mathbb{E}}_n\left[\varphi_\tau(O;\eta_0) \mid V = v\right].$$

This benchmark is not available in practice; it is used only to measure the cost of estimating the nuisance functions.

Define the conditional pseudo-outcome bias

$$b_{\widehat{\eta}}(v) = \mathbb{E}_{P_0}\left[\widehat{\varphi}_\tau(O) - \varphi_\tau(O;\eta_0) \mid \widehat{\eta}, V = v\right].$$

Equivalently, $b_{\widehat{\eta}}(v)$ is the regression function of the pseudo-outcome estimation error on V, conditional on the nuisance training sample.

A7 (Second-stage stability)**.** We say that $\widehat{\mathbb{E}}_n$ is stable at $V = v$, with respect to a distance d, if

$$\frac{\widehat{\mathbb{E}}_n\left[\widehat{\varphi}_\tau(O) \mid V = v\right] - \widehat{\mathbb{E}}_n\left[\varphi_\tau(O;\eta_0) \mid V = v\right] - \widehat{\mathbb{E}}_n\left[b_{\widehat{\eta}}(V) \mid V = v\right]}{\sqrt{\mathbb{E}\left[\left(\widehat{\mathbb{E}}_n\left[\varphi_\tau(O;\eta_0) \mid V = v\right] - \mathbb{E}_{P_0}\left[\varphi_\tau(O;\eta_0) \mid V = v\right]\right)^2\right]}} \xrightarrow{p} 0$$

whenever

$$d\{\widehat{\varphi}_\tau, \varphi_\tau(\eta_0)\} \xrightarrow{p} 0.$$

Theorem 5 (Oracle equivalence of the LMTP DR-Learner)**.** *Suppose the identification assumptions hold for both policies d_1 and d_0. Suppose the nuisance functions are estimated with cross-fitting and that Assumption A7 holds. Then*

$$\widehat{\tau}_{\mathrm{dr}}(v) - \widetilde{\tau}(v) = \widehat{\mathbb{E}}_n \left[b_{\widehat{\eta}}(V) \mid V = v \right] + o_p\{R_n^\star(v)\}.$$

Consequently,

$$\widehat{\tau}_{\mathrm{dr}}(v) - \tau(v) = \widetilde{\tau}(v) - \tau(v) + \widehat{\mathbb{E}}_n \left[b_{\widehat{\eta}}(V) \mid V = v \right] + o_p\{R_n^\star(v)\}.$$

If

$$\widehat{\mathbb{E}}_n \left[b_{\widehat{\eta}}(V) \mid V = v \right] = o_p\{R_n^\star(v)\},$$

then

$$\widehat{\tau}_{\mathrm{dr}}(v) - \widetilde{\tau}(v) = o_p\{R_n^\star(v)\}.$$

Thus the feasible DR-Learner is oracle equivalent at v.

Proof. Conditional on the nuisance training sample, decompose the estimated pseudo-outcome as

$$\widehat{\varphi}_\tau(O) = \varphi_\tau(O; \eta_0) + b_{\widehat{\eta}}(V) + e_{\widehat{\eta}}(O),$$

where

$$\mathbb{E}_{P_0}\{e_{\widehat{\eta}}(O) \mid \widehat{\eta}, V\} = 0.$$

Applying the second-stage regression estimator gives

$$\widehat{\tau}_{\mathrm{dr}}(v) = \widehat{\mathbb{E}}_n \left[\varphi_\tau(O; \eta_0) + b_{\widehat{\eta}}(V) + e_{\widehat{\eta}}(O) \mid V = v \right].$$

By Assumption A7,

$$\widehat{\tau}_{\mathrm{dr}}(v) - \widehat{\mathbb{E}}_n \left[\varphi_\tau(O; \eta_0) \mid V = v \right] = \widehat{\mathbb{E}}_n \left[b_{\widehat{\eta}}(V) \mid V = v \right] + o_p\{R_n^\star(v)\}.$$

Since

$$\widetilde{\tau}(v) = \widehat{\mathbb{E}}_n \left[\varphi_\tau(O; \eta_0) \mid V = v \right],$$

the first equation of the theorem follows. Adding and subtracting $\tau(v)$ gives the second equation. The oracle equivalence claim follows if the smoothed conditional bias is negligible relative to the oracle regression error. □

Corollary 6 (Consistency of the DR-Learner)**.** *Suppose the conditions of Theorem 5 hold. If the oracle regression is consistent for the truth, and the conditional pseudo-outcome bias decays to zero, then the oracle regression will be consistent.*

Proof. The result follows immediately from the expansion in Theorem 5. □

4.3 Second-order bias and nuisance-rate conditions

To be rewritten

5 Simulation Study

5.1 Results

6 Application

7 Discussion

8 References

References

DÍAZ, I., WILLIAMS, N., HOFFMAN, K. L. & SCHENCK, E. J. (????). Nonparametric Causal Effects Based on Longitudinal Modified Treatment Policies **118**, 846–857.

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9 Acknowledgments